

Labwork 2 Report: Get to know your machine

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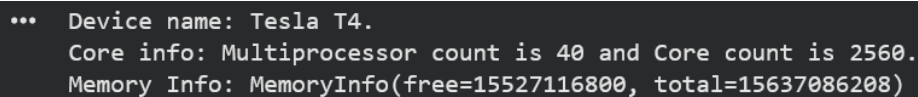
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1 Objectives & Breakdown

In this lab, we want to explore the Python Numba basic functions to recover basic information regarding our GPU.

2 Code Execution & Results

Below is the screenshot demonstrating the execution and final output of the code:

A screenshot of a terminal window with a dark background and light-colored text. The text displays the output of a program, showing GPU details for a Tesla T4. The output is as follows:

```
... Device name: Tesla T4.  
Core info: Multiprocessor count is 40 and Core count is 2560.  
Memory Info: MemoryInfo(free=15527116800, total=15637086208)
```

Figure 1: Screenshot of the program's output.

3 Sources & References

The following resources and documentation were used to complete this work:

- Numba CUDA Host API Documentation:
<https://numba.readthedocs.io/en/stable/cuda-reference/host.html#device-management>
- Stack Overflow guide that allowed me to get the numbers of cores and the multiprocessor count:
<https://stackoverflow.com/questions/63823395/how-can-i-get-the-number-of-cuda-cores-in-63833950#63833950>